All 6 exploration agents have returned. Let me synthesize their findings into one coherent report.

PROJECT OVERVIEW

GarTexHub is a B2B textile and garments marketplace platform that connects global buyers, garment factories, and buying houses into a single trust-first, governance-driven, real-time trading ecosystem. It is designed to digitize the entire textile sourcing lifecycle — from discovery and verification through negotiation, contract signing, and deal closure.

ONE-SENTENCE PROJECT DEFINITION

> GarTexHub is a trust-first B2B marketplace that connects verified garment buyers with factories and buying houses through AI-powered discovery, governed communication, electronic contracts, and CRM lead management — serving as the operating system for the global textile supply chain.

PROJECT PURPOSE

The textile sourcing industry is fragmented across email, WhatsApp, trade shows, and spreadsheets, with pervasive trust issues — ghost factories, fake registrations, anonymous buyers, and no audit trail on negotiations. GarTexHub was created to:

- **Eliminate trust gaps** through document verification, admin review, trust-risk scoring, and visible badges
- **Centralize discovery** with AI-powered semantic search across 40+ textile specification dimensions
- **Govern communication** with anti-spam policy engines, reputation scoring, and outreach controls
- **Formalize deals** with state-machine-driven workflows, e-signature contracts, and payment proof tracking
- **Enable CRM operations** with lead pipelines, SLA tracking, agent assignments, and escalation management
- **Monetize through freemium** — free core access with premium tiers unlocking advanced filters, higher volume, and priority visibility

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PROJECT VISION

GarTexHub appears designed to become the **trade operating system for the global garment and textile industry** — a single platform where an entire B2B transaction lifecycle (discovery → verification → negotiation → contract → payment → rating) occurs within a governed, auditable, AI-assisted ecosystem. The architecture supports multi-industry expansion and trade finance infrastructure (payment proofs with LC/Letter of Credit fields already exist in the data model).

TARGET USERS

User Type	Who They Are	What They Accomplish
Buyers	International fashion brands, retailers, importers sourcing garments	Publish structured requirements, discover verified factories, negotiate, sign contracts, rate counterparties
Factories	Garment manufacturers and textile mills (primarily Bangladesh-based)	List products with rich media, respond to buyer requirements, earn verified badges, build reputation
Buying Houses	Sourcing intermediaries managing networks of factories	Own partner networks, manage agent teams, run CRM lead pipelines with SLA enforcement
Agents	Operational staff assigned to handle leads within buying houses	Work assigned leads through stages, add notes, set reminders, escalate overdue deals
Owners/Admins	Platform operators and system administrators	Govern the ecosystem via dynamic admin panels, manage verification queues, enforce policies, monitor analytics

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PROBLEMS IT SOLVES

Problem	Project Response	Intended Benefit
Trust deficit — buyers can't verify factories are real	Document verification with region-specific checklists, admin review, trust-risk scoring	Reduced fraud and confident trading
Discovery fragmentation — requirements scattered across channels	Centralized feed with 40+ spec fields, OpenSearch + Qdrant semantic search	Faster, more accurate supplier matching
Communication chaos — deals happen across email, WhatsApp, phone	Real-time WebSocket chat with communication policy engine, WebRTC calls	Auditable, governed communication
Contract and compliance risk — informal verbal agreements	E-sign contracts with provider adapters, webhook retry workers, audit trails	Legally-binding digital deal execution
Dropped leads — no structured follow-up	CRM pipeline with SLA timers, agent assignments, escalations, reminders	Higher conversion rates and accountability
Spam and abuse on open platforms	Multi-language content moderation, sender reputation, outreach caps, message queuing	Maintained platform quality and user trust
Pricing confusion across currencies	Real-time FX rate normalization via Frankfurter API	Consistent cross-border price comparison
Manual admin overhead	Dynamic config-driven admin panels, automated verification, AI moderation	Reduced operational burden

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CORE CAPABILITIES

1. **Verified Marketplace Discovery** — Multi-dimensional search (OpenSearch keyword + Qdrant semantic) across products, requirements, and profiles with 37+ filter types
2. **Identity Verification** — Role/region-specific document checklists, admin review queues, duplicate detection, subscription-backed badges
3. **Real-Time Communication** — WebSocket chat with policy engine, WebRTC voice/video calls, presence tracking, conversation locking

4. **AI Intelligence** — Floating assistant, requirement extraction from natural language, chatbot qualification, content moderation, hallucination detection
5. **Contract & E-Sign** — Full contract lifecycle with Dropbox Sign integration, webhook retry, payment proof management
6. **CRM Lead Pipeline** — Lead assignment strategies, SLA timers, escalation workflows, agent workload management
7. **Governance Engine** — Policy definitions with versions, trust-risk scoring, graduated enforcement, appeals, audit trails
8. **Social Feed** — Buyer requests + company products in categorized streams with boosts, hashtags, comments, and real-time updates
9. **Monetization** — Freemium subscriptions, wallet system, paid boosts, coupons, premium-gated features
10. **Admin Control** — 12-section dynamic admin panel covering security, infrastructure, CMS, finance, media review, and file management

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FEATURE CATALOG

Authentication & Identity

- **Multi-role system** (buyer, factory, buying_house, agent, owner, admin) — tailored experiences per actor type
- **JWT authentication** — stateless API security with 12h tokens
- **WebAuthn passkeys** — passwordless biometric authentication for enterprise security
- **Onboarding wizard** — guided 3-step setup for new users

Verification & Trust

- **Document verification** — trade licenses, TIN/EIN/EORI, bank proofs, export certificates per region
- **Duplicate detection** — 8-signal weighted scoring across name, country, address, website, email, phone, identifier, logo
- **Trust-risk scoring** — multi-signal composite from verification recency, disputes, messaging patterns, contract breaches
- **Verification review workflow** — admin pipeline with pending/approved/rejected/incomplete/duplicate statuses

Marketplace & Search

- **OpenSearch full-text search** — faceted filtering across category, MOQ, price, certifications, lead time, GSM, incoterms
- **Qdrant semantic search** — vector similarity for natural-language queries ("lightweight summer fabrics for kids")
- **Search presets** — save, share, and reuse complex filter configurations
- **Search alerts** — persistent subscriptions notified when new results match saved queries
- **Industry pages** — dedicated landing pages per textile vertical
- **Currency normalization** — real-time FX rates enabling cross-border price comparison

Feed & Content

- **Social feed** — interleaved buyer requests, company products, and user posts with categories
- **Feed post management** — markdown, media, hashtags, mentions, CTA buttons, link previews
- **Feed boosts** — paid visibility promotion with quality-based ranking and abuse detection
- **Shared post pages** — public-facing shareable links for external distribution

Communication

- **Thread-based messaging** — match-linked conversations with attachments, read receipts, AI summaries
- **Communication policy engine** — outbound caps, cooldown periods, keyword risk scoring (EN/BN/ZH/ES), sender reputation
- **Message queuing** — high-risk messages enter review queue with priority scoring
- **Conversation locking** — agent-level exclusive claim on buyer request threads
- **WebRTC calls** — peer-to-peer video/audio with STUN/TURN, call recording, audit trails
- **Presence system** — online/offline status with last-seen timestamps

AI & Intelligence

- **Floating AI assistant** — platform-aware streaming chatbot (OpenCode + Gemini) with knowledge base
- **Requirement extraction** — LLM-powered structured extraction from free-text buyer messages
- **Chatbot qualification** — automated buyer walk-through of specification questions
- **AI auto-reply** — configurable automated responses with handoff thresholds and hallucination detection
- **Content moderation** — Python YOLO pipeline for image/video safety scoring (SAFE/QUESTIONABLE/HARAM/HIGH_RISK)
- **Search reranking** — cross-encoder relevance refinement post-retrieval

Contracts & E-Sign

- **Document lifecycle** — draft → pending signature → signed → archived with artifact metadata
- **E-signature integration** — Dropbox Sign adapter with webhook callbacks, nonce verification, retry workers
- **Payment proof management** — upload, review, approve/reject with banking metadata
- **Contract vault** — centralized repository with artifact audit, banking references, workflow summaries

CRM & Leads

- **Lead pipeline** — full lifecycle: new → contacted → negotiating → sample_sent → order_confirmed → closed
- **Lead assignment** — round-robin, least-loaded, skill-based strategies with agent capacity tracking
- **SLA monitoring** — time-based targets per stage with automatic escalation on breach
- **Lead notes & reminders** — team collaboration with background worker sweep and email notifications
- **CRM summary panel** — 360-degree relationship view with interaction history, calls, documents

Ratings & Feedback

- **Bidirectional ratings** — score breakdown (1-5 stars), milestone tracking, auto-rating after configurable days
- **Order completion certification** — automated certification triggered by contract + call completion

Organization Management

- **Member management** — invitation, permission matrix, seat caps by subscription tier
- **Organization settings** — team config, AI settings, operational preferences
- **Agent sub-IDs** — tracked sub-identities for agents within organizations
- **Join requests** — request-to-join workflow with approval/rejection and reminder sweeps

Monetization

- **Subscription system** — free/premium tiers with auto-renewal and plan history
- **Entitlement engine** — 40+ feature flags per role/plan (premium buyers get advanced filters, priority placement)
- **Wallet system** — USD-denominated with credit/debit, restricted balance, transaction history
- **Boost system** — paid feed visibility with quality ranking and abuse detection
- **Coupons** — promotional codes with redemption limits and expiry

Analytics & Insights

- **Event tracking** — structured ingestion with taxonomy, idempotency, context enrichment
- **Analytics dashboard** — core metrics, conversion funnels, search trends, segment analysis
- **Analytics governance** — privacy-preserving with minimum cohort sizes, field-level denials
- **Analytics export** — governed data extraction with audit logging and access policies
- **Product view tracking** — deduplication windows, geo-enrichment, event integration

Admin & Governance

- **12-section admin panel** — Home, AI Config, CMS, Platform Config, Finance, Infrastructure, Media Review, Network, Security, Server, File Explorer
- **Dynamic configuration** — database-driven UI modules, feature flags, plan limits, pricing, policies
- **Governance engine** — policy registry with versions, simulation, enforcement history, appeals, monthly reports
- **Dual-confirm admin actions** — two-step confirmation for destructive operations
- **Step-up authentication** — re-authentication for sensitive admin operations
- **Audit logging** — immutable trail of all admin actions with actor, timestamp, entity, IP

Real-Time Infrastructure

- **WebSocket bus** — notification, feed, chat, call, and presence events
- **SSE feed stream** — server-sent events for live marketplace updates
- **Presence tracking** — online/offline with heartbeat-based liveness

Monitoring & Observability

- **Log TUI dashboard** — 23+ widget terminal UI with live streaming, inspector, sidebar, filter bar, status bar, themes
- **Structured logging** — 6 levels, 17 categories, plugin-based parsers, burst detection
- **Request flow waterfall** — Chrome DevTools-style request lifecycle visualization
- **Syslog server** — TLS-enabled RFC 5424 for external system log ingestion

Media Processing

- **Image processing** — Sharp + ImageMagick fallback for compression and format normalization
- **Video processing** — ffprobe validation + ffmpeg compression in background queues
- **AI moderation** — YOLO/NudeNet/PaddleOCR pipeline for content safety scoring

Infrastructure

- **PostgreSQL** via Prisma (80+ models, 19 migrations)
- **Redis** optional caching with graceful degradation

- **OpenSearch** full-text and faceted search
- **Qdrant** vector database for semantic search
- **Currency/FX service** with hourly rate refresh
- **Geo service** with IP-based geolocation and caching

MAJOR SYSTEM AREAS

System Area	Purpose	Why It Exists
User Identity & Access	Authenticate users, enforce roles, manage organizations	Every interaction must be attributable and role-appropriate
Marketplace Discovery	Search, match, and connect buyers with factories	The core value proposition — finding the right trade partners
AI Intelligence	Assist, extract, moderate, and moderate content	Reduce friction and maintain quality at scale
Communication	Chat, calls, presence, policy enforcement	Deals close through conversation, not just search
Deal Lifecycle	State-machine tracking from discovery to closure	Every deal must have an auditable progression
CRM & Operations	Lead pipelines, SLA tracking, agent management	Buying houses need enterprise sales tools
Trust & Governance	Verification, scoring, enforcement, appeals	Trust is the currency of B2B trade
Contracts & E-Sign	Document creation, digital signing, payment proofs	Formal agreements are legally required
Financial	Wallets, subscriptions, boosts, coupons	Sustainable revenue through freemium monetization
Admin & Control	Dynamic panels, CMS, infrastructure, security	The platform must be governable at scale
Observability	Logging, monitoring, analytics, TUI dashboard	Production operations require visibility

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USER EXPERIENCE & USE CASES

Primary Use Cases

1. **Buyer discovers a factory** → searches with semantic/keyword filters → views verified profile → initiates chat → negotiates → signs contract → rates experience
2. **Factory lists products** → earns verified badge → receives smart match notifications → responds to buyer requirements → closes deals
3. **Buying house runs operations** → manages partner network → assigns agents to leads → monitors SLAs → escalates overdue deals → reviews pipeline analytics

Secondary Use Cases

4. **New user onboards** → registers with role selection → guided wizard → lands on feed → starts exploring
5. **Agent works a lead** → receives assignment → contacts buyer → adds notes → sets reminders → escalates if needed → closes or hands off
6. **Admin governs platform** → reviews verification queue → monitors governance reports → manages policies → handles appeals

Advanced Use Cases

7. **Premium subscriber** → unlocks 24 advanced filters, 400 daily searches, priority placement, dedicated support
8. **Organization owner** → configures team permissions, manages partner factories, reviews CRM analytics, runs the full lead pipeline
9. **Platform operator** → uses TUI dashboard for real-time monitoring, manages infrastructure via admin panel, reviews audit trails

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MAJOR WORKFLOWS

Discovery → Deal Closure:

> Buyer posts requirement → AI extracts structured specs → matching engine scores factory fit → factories receive notifications → interested factory initiates chat → negotiation via governed messaging → escalation to video call → contract drafted and e-signed → payment proof uploaded → order certified → both parties rate each other

Verification Pipeline:

> User uploads documents per role/region checklist → duplicate detection scans → admin reviews submission → approved/rejected with reason → verified badge appears on profile → subscription-linked validity tracked

Lead Management:

> Owner assigns lead to agent → agent receives notification with SLA timer → agent works through stages (new → contacted → negotiating → sample → confirmed → closed) → notes and reminders tracked → SLA breach triggers escalation → owner reviews pipeline analytics

Governance Enforcement:

> Message scored for policy risk → low risk: delivered → medium risk: queued for review → high risk: blocked → sender reputation updated → repeated violations trigger graduated enforcement (warning → throttle → lock) → user can appeal → admin reviews and resolves

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INTEGRATIONS & EXTERNAL SYSTEMS

Integration	Why It Exists	Capability It Enables
PostgreSQL (Aiven)	Primary relational database	Persistent storage for all platform entities
OpenSearch (Aiven)	Full-text and faceted search	Multi-dimensional product/requirement discovery
Qdrant	Vector database	Semantic similarity search for natural-language queries
Redis	Caching layer	Performance optimization and rate limiting
Ollama	Local LLM inference	Zero-cost AI for development and on-premise
OpenRouter	Cloud LLM fallback	Higher-quality inference when local models unavailable
Google Gemini	AI provider	Alternative LLM path for assistant and extraction
HuggingFace	Embeddings + reranking	Production vector search and result refinement
OpenCode	Embedded AI server	Per-user streaming assistant with session management

Integration	Why It Exists	Capability It Enables
Frankfurter API	Exchange rates	Multi-currency price normalization
Dropbox Sign	E-signature provider	Digital contract execution
Gmail API / SMTP	Email delivery	Transactional notifications and password resets
WebRTC (STUN/TURN)	Video/audio calls	Peer-to-peer communication between trade partners
Render	Production hosting	Managed cloud deployment with auto-deploy
Electron	Desktop wrapper	Native desktop application distribution
Python YOLO Pipeline	Content moderation	Image/video safety scoring for uploads

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AI / AUTOMATION CAPABILITIES

AI exists in the platform to:

- Extract structured textile specifications from free-form buyer text (requirement extraction)
- Qualify buyers through automated chatbot conversations (qualification bot)
- Assist users with platform navigation and sourcing best practices (floating assistant)
- Moderate uploaded content for safety and policy compliance (YOLO-based image moderation)
- Summarize long conversation threads for quick context recovery (conversation summaries)
- Detect hallucinations in AI-generated content before it enters production (hallucination detection)
- Rerank search results for improved relevance (cross-encoder reranking)
- Auto-reply in buyer-supplier conversations with configurable handoff thresholds (auto-reply)

Automation exists for:

- Lead reminder sweeps via background workers
- E-signature webhook retry with exponential backoff
- Image and video compression in background queues
- Currency rate refresh every hour
- Partner free-tier enforcement
- Search alert notification when new results match saved queries
- Graduated enforcement actions based on trust-risk scores

DATA & INFORMATION PURPOSE

The platform manages:

- **User identity data** — profiles, organizations, roles, permissions, verification documents
- **Marketplace data** — products (40+ attributes), buyer requirements (40+ fields), matches, search history
- **Communication data** — messages, call recordings, presence, read receipts, policy decisions
- **Contract data** — document lifecycle, signatures, payment proofs, certification status
- **CRM data** — leads, assignments, notes, reminders, SLA timers, escalations, interaction logs
- **Financial data** — wallet balances, subscription history, boost purchases, coupon redemptions
- **Governance data** — policies, enforcement actions, appeals, trust-risk evaluations, audit logs
- **Analytics data** — events, search trends, conversion funnels, platform metrics

All data serves the core mission: enabling trusted, efficient, auditable B2B textile trade.

SECURITY & TRUST PURPOSE

- **Authentication** — JWT with 12h expiry, issuer/audience pinning, WebAuthn passkeys for passwordless login
- **Authorization** — Role-based access with permission matrices and org-level scoping
- **Verification** — Document-based identity verification with region-specific requirements and admin review
- **Communication governance** — Anti-spam policy engine with multilingual detection, reputation scoring, outreach caps
- **Admin security** — IP allowlisting, device fingerprinting, MFA step-up, dual-confirmation for destructive actions
- **Audit trails** — Immutable logging of all admin actions, workflow transitions, and governance enforcement
- **Content safety** — YOLO-based image moderation, SVG sanitization, XSS prevention
- **Rate limiting** — Per-endpoint throttling with role-based limits
- **Zero-trust infrastructure** — Encrypted backups, tamper-proof logging, SSL certificate management

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ADMINISTRATION & MANAGEMENT PURPOSE

The admin panel exists to give platform operators complete control without leaving the application:

- **Dynamic configuration** — all platform parameters (feature flags, pricing, policies, integrations) are database-driven and changeable at runtime
- **Verification queue** — review and adjudicate identity submissions
- **Governance operations** — define policies, simulate impacts, manage enforcement and appeals
- **Infrastructure management** — firewall rules, packages, cron jobs, SSH keys, SSL certs, process monitoring
- **Content management** — article/page creation, media management, theme configuration, SEO
- **Financial oversight** — subscription management, wallet operations, refund processing
- **Security posture** — zero-trust policies, device allowlists, incident tracking, immutable backups

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PROJECT SCOPE

GarTexHub is a **multi-purpose B2B marketplace platform** — not a simple tool or single-purpose application. It combines:

- **Marketplace** features (product listings, buyer requirements, semantic matching)
- **Communication platform** (real-time chat, video calls, presence)
- **CRM system** (lead pipelines, SLA tracking, agent management)
- **E-signature platform** (contract creation, digital signing, audit trails)
- **Governance engine** (policy management, trust scoring, enforcement)
- **Analytics platform** (event tracking, dashboards, export)
- **AI assistant** (streaming chatbot, requirement extraction, moderation)
- **Admin control center** (12-section dynamic panel)

The scope covers the entire B2B textile sourcing lifecycle from discovery through contract closure. It is specialized for the textile/garment industry but architecturally extensible to other B2B verticals.

CURRENT PROJECT STATE

Indicator	Assessment
Version	0.0.0-beta
Database	80+ models, 19 migrations — mature schema
Backend	96 services, 64 controllers, 58 route modules — extensive

Indicator	Assessment
Frontend	58 page components, React 19 + Vite 6
Tests	50+ unit, 10+ integration, 2 e2e — moderate coverage
AI	5 providers integrated, multiple use cases active
Payment	Simulated wallet/billing — no real payment gateway
Deployment	Live on Render, Docker for local dev
Documentation	260+ generated doc files
Commit history	644 commits — actively developed

Classification: Actively developed beta with production-oriented architecture. Core features are implemented and functional. Payment processing is simulated. Some infrastructure cleanup remains (committed secrets, dual lockfiles).

DECLARED INTENT VS OBSERVED CAPABILITY

Area	Declared Intent	Observed Capability	Confidence
B2B marketplace	Connect buyers, factories, buying houses	Fully implemented with 6 roles, 80+ models	High
Verification system	Document-based identity verification	Full implementation with region-specific checklists, admin review	High
AI assistant	Floating streaming chatbot	OpenCode + Gemini integration with WebSocket streaming	High
Semantic search	Qdrant vector search	Implemented but may not be active in production	Medium
E-sign contracts	Dropbox Sign integration	Full implementation with retry workers	High

Area	Declared Intent	Observed Capability	Confidence
Real payment processing	Wallet-based billing	Simulated only — no real payment gateway	Low
Mobile app	Planned	Not implemented — web responsive only	Low
SMS notifications	Placeholder in config	No implementation found	Low
Slack integration	Placeholder in config	No implementation found	Low

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MAJOR STRENGTHS OF THE CONCEPT

- **Comprehensive lifecycle coverage** — from discovery through contract closure in one platform
- **Trust-first architecture** — verification, trust scoring, and governance are first-class, not afterthoughts
- **AI-native design** — AI is woven into search, extraction, moderation, assistance, and qualification
- **Multi-dimensional search** — OpenSearch + Qdrant + embeddings + reranking for precise discovery
- **Communication governance** — sophisticated anti-spam and reputation system prevents platform degradation
- **Enterprise CRM** — lead pipelines with SLA tracking and escalation rival standalone CRM tools
- **Dynamic administration** — database-driven admin UI that changes without code deployments
- **Production observability** — custom TUI dashboard with real-time monitoring, burst detection, and request flow analysis
- **Multi-deployment** — web, desktop (Electron), Docker, and cloud (Render) targets
- **Freemium monetization** — clear value ladder from free core to premium intelligence

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PROJECT SUMMARY

GarTexHub is a B2B textile marketplace platform that provides end-to-end sourcing capabilities for the global garment industry. It connects verified buyers, factories, and buying houses through AI-powered discovery, governed communication, electronic contracts, and CRM lead management. The platform addresses trust, fragmentation, and operational complexity in textile sourcing by combining marketplace mechanics, real-time communication, governance enforcement, and AI assistance into a single, auditable ecosystem. With 80+ database models, 96 services, 5 AI providers, and a custom monitoring dashboard, it represents a substantial full-stack application currently in active beta development.

EXECUTIVE DESCRIPTION

GarTexHub is a trust-first B2B marketplace platform engineered for the global garment and textile industry. It connects international buyers with verified factories and buying houses through AI-powered semantic search, real-time governed communication, electronic contract signing, and enterprise CRM lead management. The platform combines marketplace discovery, identity verification, communication policy enforcement, contract lifecycle management, and analytics into a single operating system for textile sourcing — where every interaction is auditable, every participant is verifiable, and every deal is tracked from first contact through closed order. Built with React, Express, Prisma, and multiple AI providers, GarTexHub deploys to web, desktop, and cloud environments.

VERY SHORT DESCRIPTIONS

1-line: GarTexHub is a trust-first B2B marketplace connecting verified garment buyers with factories through AI-powered discovery, governed communication, and electronic contracts.

2–3 sentences: GarTexHub is a B2B textile marketplace that digitizes the entire sourcing lifecycle — from requirement posting and semantic search through negotiation, contract signing, and deal closure. It combines identity verification, AI assistance, governed communication, and CRM lead management into a single platform for the global garment industry.

Short paragraph: GarTexHub is a full-stack B2B marketplace platform purpose-built for the global garment and textile industry. It connects buyers, factories, and buying houses through AI-powered discovery, real-time communication with anti-spam governance, electronic contract signing, and enterprise CRM tools. The platform addresses trust deficits, fragmented communication, and operational complexity in textile sourcing by providing a governed, auditable ecosystem where every interaction — from first search to signed contract — is tracked, verified, and scored.

PROJECT TAGS / CONCEPT KEYWORDS

B2B marketplace, textile industry, garment sourcing, trust verification, AI-powered search, semantic search, e-signature, CRM pipeline, governance engine, real-time communication, WebRTC, WebSocket, contract management, lead management, SLA tracking, content moderation, OpenSearch, Qdrant, freemium SaaS, React, Express, Prisma, PostgreSQL

UNKNOWN / UNCERTAIN AREAS

- **Qdrant production status** — provisioned locally but production Render deployment points to localhost, suggesting semantic search may not be live in production
- **Real payment processing timeline** — wallet is simulated; no Stripe or payment gateway integration found
- **Mobile app plans** — no React Native or mobile-specific code found despite responsive web design
- **Email verification** — email sending exists but "not yet wired" per README
- **SMS and Slack integrations** — placeholders in config with no implementation
- **Multi-industry expansion** — architecture is flexible but currently textile-specific

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EVIDENCE NOTES

- **Documentation:** README.md, docs/PROJECT_DOCS.md, AGENTS.md, .env.example provide strong declared intent
- **Database schema:** 80+ Prisma models confirm extensive data architecture
- **Directory structure:** server/services/, server/routes/, src/pages/ confirm feature breadth
- **Package.json:** Dependencies confirm AI (OpenCode SDK, Google GenAI), search (Qdrant, OpenSearch), and real-time (ws) integrations
- **Docker/Render config:** Deployment files confirm production architecture
- **Test suite:** 50+ tests confirm quality infrastructure
- **644 commits:** Development history confirms active, sustained development
- **WhatsApp stakeholder chats:** Confirm target market (Bangladeshi factories + international buyers) and design philosophy (LinkedIn-style, premium, corporate)

FINAL PROJECT UNDERSTANDING

GarTexHub is ultimately trying to solve the **trust and efficiency crisis in global textile sourcing**. The complete set of features exists together because textile trade requires every one of them to function as a coherent whole: buyers need to discover verified suppliers (marketplace + search), verify their legitimacy (verification + trust scoring), communicate securely (messaging + policy engine), formalize agreements (contracts + e-sign), manage follow-through (CRM + SLA), and do all of this within a governed, auditable ecosystem (governance + admin + analytics). The AI layer reduces friction at every step — extracting requirements from messy text, qualifying buyers through conversation, moderating content, and assisting users across the platform. The freemium model makes it accessible while premium features drive sustainable revenue. Together, these capabilities create not just a marketplace,

but a **trade operating system** where the entire lifecycle of a B2B textile transaction — from first search to signed contract to post-deal rating — occurs within a single, trust-scored, AI-assisted platform.